

Nectar Data Scientist Challenge

Position

Role: Data Scientist

Experience: 1–3 Years

Location: Coimbatore, Tamil Nadu, India

About Nectar

Nectar is an AI-first product company building intelligent platforms that transform how enterprises leverage AI, automation, and connected technologies.

Our platform combines Artificial Intelligence, IoT, Agentic AI, and cloud-native technologies to deliver intelligent, scalable, and secure experiences for customers worldwide.

As a Data Scientist at Nectar, you will work on real-world challenges involving connected assets, sensor telemetry, predictive analytics, and operational intelligence.

This assessment is designed to evaluate your analytical thinking, machine learning expertise, and ability to convert IoT data into actionable product capabilities.

Challenge Overview

You are working on Nectar's Intelligent Facilities Platform.

A customer operates multiple commercial buildings, each containing several interconnected assets such as:

- HVAC Systems
- Chillers
- Pumps
- Air Handling Units (AHUs)
- Energy Meters
- Environmental Sensors

Thousands of sensors continuously stream telemetry data from these assets.

Your objective is to analyze historical IoT data, understand asset relationships, identify operational inefficiencies, and build intelligent capabilities that can improve reliability, reduce downtime, and optimize energy consumption.

1. Sensor Telemetry

Historical sensor readings collected from connected assets.

Column	Description
timestamp	Timestamp of sensor reading
site_id	Unique site identifier
building_id	Building identifier
asset_id	Unique asset identifier
temperature	Temperature reading (°C)
humidity	Humidity (%)
pressure	Pressure reading
vibration	Equipment vibration level
power_consumption	Energy consumption (kWh)
occupancy_count	Number of occupants
operating_mode	Cooling/Heating/Idle
fault_flag	Historical fault indicator (0/1)

2. Asset Metadata

Information describing deployed assets.

Column	Description
asset_id	Unique asset identifier
site_id	Site identifier
asset_name	Asset name
asset_type	Chiller/AHU/Pump/etc.
manufacturer	Manufacturer
installation_date	Installation date
capacity	Rated capacity
parent_asset_id	Parent asset relationship

3. Asset Connectivity

Represents relationships between assets.

Column	Description
source_asset_id	Parent asset
target_asset_id	Connected asset
connection_type	Supplies/Controls/Monitors
relationship_strength	Connectivity weight

Tasks

Task 1: Exploratory Data Analysis (EDA)

Perform a detailed exploratory analysis on the telemetry data.

Your analysis should:

- Understand data distributions.
- Identify missing values and anomalies.
- Discover temporal patterns.
- Analyze asset behavior across sites.
- Compare performance across asset types.
- Identify factors influencing equipment failures.
- Identify factors affecting energy consumption.

Deliverables

Provide:

- Visualizations
 - Statistical summaries
 - Key observations
 - Business insights
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Task 2: Predictive Maintenance

Develop a machine learning model to predict whether an asset is likely to fail within the next 24 hours.

The model should help operations teams proactively schedule maintenance activities.

Expected Activities

- Data preprocessing
- Feature engineering
- Handling missing values
- Model selection

- Model training
- Model evaluation
- Error analysis

Suggested Algorithms

- Random Forest
- XGBoost
- LightGBM
- CatBoost
- Any appropriate approach

Evaluation Metrics

- Precision
- Recall
- F1 Score
- ROC-AUC

Explain

- Why the selected model was chosen.
 - Important features driving predictions.
 - Potential business impact.
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Task 3: Energy Consumption Forecasting

Develop a forecasting model to predict energy consumption for the next 24 hours for each building.

The goal is to support proactive energy optimization.

Suggested Techniques

- Prophet
- ARIMA
- XGBoost
- LSTM
- Temporal Fusion Transformer

- Any suitable approach

Evaluation Metrics

- MAE
- RMSE
- MAPE

Expected Output

- Forecast plots
 - Model evaluation
 - Key insights
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Task 4: Anomaly Detection

Design an anomaly detection framework capable of identifying abnormal equipment behavior.

Examples include:

- Sudden power spikes
- Excessive vibration
- Sensor drift
- Abnormal temperature patterns
- Equipment degradation

Suggested Techniques

- Isolation Forest
- Autoencoders
- Statistical Thresholding
- Clustering
- Change Point Detection

Deliverables

- Anomaly detection methodology
- Visualizations
- Identified anomalies

- Business recommendations
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Task 5: Multi-Asset Connectivity Analysis

Each customer site contains multiple interconnected assets.

For example:

Site A

- Chiller-01
 - AHU-01
 - AHU-02
- Pump-01
 - Flow Sensor-01
 - Energy Meter-01

Your objective is to model and analyze relationships between assets.

Requirements

Build Asset Hierarchy

Create a hierarchical representation of assets within each site.

Examples:

- Site → Building → Floor → Asset
 - Chiller → AHU → Sensor
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Connectivity Analysis

Analyze relationships among assets and answer:

- Which assets are connected?
 - Which assets depend on each other?
 - Which assets are isolated or disconnected?
 - What downstream assets are affected if a parent asset fails?
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Failure Impact Analysis

Assume a critical asset fails.

Examples:

- Chiller failure
- Pump failure

Determine:

- Impacted downstream assets
 - Potential operational impact
 - Recommended mitigation strategies
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Data Quality Assessment

Identify:

- Missing relationships
 - Duplicate connections
 - Orphan assets
 - Invalid parent-child mappings
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Expected Deliverables

Provide:

- Asset hierarchy visualization
 - Dependency graph/network visualization
 - Connectivity analysis report
 - Failure propagation analysis
 - Data quality findings
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Bonus

Use graph-based techniques such as:

- GraphQL

Implement example queries such as:

- Show all assets connected to Chiller-01.
 - Identify all downstream assets impacted by AHU-03 failure.
 - List all assets under Site A.
 - Find isolated assets with no parent or child relationships.
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Bonus Challenge (Optional)

Complete any one of the following.

Option A: Interactive Dashboard

Develop an interactive dashboard using:

- Streamlit
- Power BI
- Tableau

Dashboard should include:

- Site overview
 - Asset health status
 - Failure predictions
 - Energy trends
 - Anomaly alerts
 - Asset connectivity visualization
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Model Deployment

Deploy one machine learning model as an API using:

- FastAPI
- Flask

The API should expose prediction endpoints.

Example:

POST /predict_failure

Input:

Asset telemetry data

Output:

Failure probability

Deliverables

Please submit:

1. Source Code (GitHub Repository preferred)
 2. Jupyter Notebook(s)
 3. README containing:
 - Setup instructions
 - Architecture overview
 - Assumptions
 - Design decisions
 4. Report or Presentation (Maximum 5 pages/slides)
 5. Optional deployment/demo link
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Evaluation Criteria

Criteria	Weightage
Problem Understanding	10%
Exploratory Analysis	15%
Feature Engineering	10%
Predictive Modeling	20%
Forecasting	10%
Anomaly Detection	10%

Asset Connectivity Analysis 15%

Code Quality & Documentation 5%

Communication & Presentation 5%

Submission Timeline

Please submit your completed assignment within **5 calendar days** from the date of receiving the challenge.

Important Notes

- Clearly document assumptions wherever applicable.
- Focus on explainability and business impact.
- Production-quality code and reproducibility are highly valued.
- Creativity and experimentation are encouraged.

We are interested not only in the final solution but also in understanding your thought process, trade-offs, and engineering decisions.

We look forward to reviewing your submission.

Team Nectar